

The Zeta–Weil Form and Nonunital GNS Transfer

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Abstract

The arithmetic terms of the zeta explicit formula define a Hermitian pairing on a nonunital convolution algebra. Weil’s theorem identifies its positivity with the Riemann hypothesis. A local Gaussian calculation proves a further restriction: the full Weil functional admits no positive unitization of finite mass and no realization by a finite-norm vector on an invariant $*$ -representation domain. The same calculation gives positive Gaussian subspaces of every finite dimension at sufficiently small scales. We construct the common graph completion of nonunital GNS multipliers and prove its exact domain and transfer properties. Closed quadratic forms for individual multipliers need not have invariant domains. Their common domain over the whole algebra does. An independent positive arithmetic functional remains open, with its possible representation constrained by the Gaussian divergence.

1 The exact zeta–Weil form

The decisive question is whether the arithmetic expression below is a squared norm for a reason independent of the location of the zeros. Such a norm could construct a Hilbert space from test functions. It need not come from a distinguished vector representing the functional. The Gaussian calculation in Section 6 will exclude that stronger realization. We first fix the arithmetic expression completely.

All integrals on $\mathbb{R}$ use Lebesgue measure. Set

$$\mathcal{A}_\zeta = \left\{ f \in C^\infty(\mathbb{R}) : \begin{array}{l} \text{there is an } \epsilon > 0 \text{ such that} \\ e^{(1/2+\epsilon)|u|} f^{(k)}(u) \in L^1(\mathbb{R}, du) \quad \text{for every } k \geq 0 \end{array} \right\}.$$

For $f, g \in \mathcal{A}_\zeta$, define

$$(f \star g)(u) = \int_{\mathbb{R}} f(v)g(u-v) dv, \quad f^*(u) = \overline{f(-u)}, \quad (1.1)$$

and use the Fourier convention

$$\widehat{f}(z) = \int_{\mathbb{R}} f(u)e^{izu} du, \quad f(u) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{f}(t)e^{-itu} dt. \quad (1.2)$$

The inverse formula in (1.2) is asserted for real u and integrates with respect to dt .

Proposition 1.1. *The space $\mathcal{A}_\zeta$ is a commutative nonunital $*$ -algebra under (1.1). If f has exponent ϵ , then $\widehat{f}$ is holomorphic on $|\operatorname{Im} z| < 1/2 + \epsilon$. In every common strip of holomorphy,*

$$\widehat{f \star g}(z) = \widehat{f}(z)\widehat{g}(z), \quad \widehat{f^*}(z) = \overline{\widehat{f}(\bar{z})}. \quad (1.3)$$

In particular,

$$\widehat{f^* \star f}(t) = |\widehat{f}(t)|^2 \quad (t \in \mathbb{R}). \quad (1.4)$$

Proof. Choose $0 < \delta < \min(\epsilon_f, \epsilon_g)$ and put $a = 1/2 + \delta$. The inequality $e^{a|u|} \leq e^{a|v|}e^{a|u-v|}$ and the weighted L^1 convolution inequality show that

$$e^{a|\cdot|}(f \star g)^{(k)} = e^{a|\cdot|}(f^{(k)} \star g) \in L^1(\mathbb{R})$$

for every k . The same estimates justify Fubini's theorem in (1.3); the assertion for the involution follows by $u \mapsto -u$. Weighted integrability of all derivatives gives Fourier inversion and rapid decay of $\widehat{f}$ on each closed substrip.

If $e \in \mathcal{A}_\zeta$ were a convolution identity, convolution with a Gaussian and (1.3) would give $\widehat{e}(t) = 1$ for every real t . This contradicts the Riemann–Lebesgue lemma, since $e \in L^1(\mathbb{R})$. $\square$

Let

$$\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2), \quad \psi(s) = \frac{\Gamma'(s)}{\Gamma(s)}.$$

Thus the archimedean normalization is fixed by

$$2 \operatorname{Re} \frac{\Gamma'_{\mathbb{R}}}{\Gamma_{\mathbb{R}}} \left(\frac{1}{2} + it \right) = \operatorname{Re} \psi \left(\frac{1}{4} + \frac{it}{2} \right) - \log \pi. \quad (1.5)$$

Write $\Lambda_{\text{vM}}(n) = \log p$ when $n = p^m$ with p prime and $m \geq 1$, and $\Lambda_{\text{vM}}(n) = 0$ otherwise. Define the linear functional $L_\zeta : \mathcal{A}_\zeta \rightarrow \mathbb{C}$ by

$$\begin{aligned} L_\zeta(g) &= \widehat{g}(i/2) + \widehat{g}(-i/2) && \text{(polar)} \\ &+ \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{g}(t) \left[\operatorname{Re} \psi \left(\frac{1}{4} + \frac{it}{2} \right) - \log \pi \right] dt && \text{(archimedean)} \\ &- \sum_{n \geq 2} \frac{\Lambda_{\text{vM}}(n)}{\sqrt{n}} (g(\log n) + g(-\log n)) && \text{(prime powers)}. \end{aligned} \quad (1.6)$$

The centered Mellin parameter in these conventions is

$$\widehat{g}(-i(s - \tfrac{1}{2})) = \int_0^\infty g(\log x) x^{s-1/2} \frac{dx}{x}. \quad (1.7)$$

Equation (1.7) is the logarithmic form of Weil's Mellin convention [2, p. 256]; functions in $\mathcal{A}_\zeta$ satisfy his conditions (A) and (B) on p. 259.

Proposition 1.2. *Every term in (1.6) converges, and*

$$L_\zeta(g^*) = \overline{L_\zeta(g)}. \quad (1.8)$$

If ρ runs over the nontrivial zeros of $\zeta(s)$, counted with multiplicity, then

$$L_\zeta(g) = \sum_{\rho} \widehat{g}(-i(\rho - \tfrac{1}{2})), \quad (1.9)$$

and the sum is absolutely convergent.

Proof. For a suitable $\epsilon > 0$, weighted $W^{1,1}$ estimates on each half-line give

$$|g(u)| \leq C e^{-(1/2+\epsilon)|u|}.$$

The prime-power series is consequently dominated by $C \sum_{n \geq 2} \Lambda_{\text{vM}}(n) n^{-1-\epsilon}$. Repeated integration by parts makes $\widehat{g}(t)$ rapidly decreasing on $\mathbb{R}$, whereas the bracket in (1.6) is $O(\log(2 + |t|))$. This proves convergence of the local terms.

For real t , the bracket is real and even, while $\widehat{g}^*(t) = \overline{\widehat{g}(t)}$; the two polar values and the two prime-power values are interchanged by the involution. This proves (1.8).

Formula (1.9) is the $k = \mathbb{Q}$, trivial-character specialization of Weil's formula (11), not a formula printed there verbatim in the normalization (1.2)–(1.7); formula (11) and the ensuing positivity lemma are on p. 262 of [2]. Repeated integration by parts is uniform for $|\text{Im } z| \leq 1/2$, and the standard zero count is $O(T \log T)$. Hence the zero sum is absolute for the present test class. $\square$

The calculation that controls the question is now immediate.

Theorem 1.3 (Weil's positivity criterion). *With the conventions above,*

$$\text{RH} \iff L_\zeta(f^* \star f) \geq 0 \quad \text{for every } f \in \mathcal{A}_\zeta. \quad (1.10)$$

Under RH, writing each zero as $\rho = 1/2 + i\gamma_\rho$, one has the exact norm calculation

$$L_\zeta(f^* \star f) = \sum_{\rho} |\widehat{f}(\gamma_\rho)|^2. \quad (1.11)$$

Proof. If RH holds, substitute $\rho = 1/2 + i\gamma_\rho$ in (1.9) and use (1.4); this gives (1.11). Conversely, an off-line zero is separated by Weil's polynomial–Gaussian construction [2, p. 263], which makes the zero side of (1.9) negative on a convolution square. The inverse Fourier transforms used there are polynomial–Gaussian functions and therefore belong to $\mathcal{A}_\zeta$. Thus positivity on the class specified here implies RH. $\square$

The right side of (1.11) explains positivity conditional on RH; it cannot serve as an independent construction of the norm, because both the real parameters γ_ρ and their existence in that form use the asserted zero placement.

2 Algebraic GNS without a unit

Let $\mathfrak{A}$ be a complex $*$ -algebra, with no norm or unit assumed, and let $W : \mathfrak{A} \rightarrow \mathbb{C}$ be positive:

$$W(a^*a) \geq 0 \quad (a \in \mathfrak{A}).$$

Theorem 2.1 (Nonunital algebraic GNS). *Put*

$$\langle a, b \rangle_W = W(b^*a), \quad \mathcal{N}_W = \{a : W(a^*a) = 0\}.$$

Then $\mathcal{N}_W$ is a left ideal and

$$D_W = \mathfrak{A}/\mathcal{N}_W, \quad \pi_W(a)[b] = [ab], \quad (2.1)$$

is a $$ -representation on a pre-Hilbert space. Regard D_W as a common invariant dense domain in its Hilbert completion $\mathcal{H}_W$. For every $a \in \mathfrak{A}$,*

$$\pi_W(a^*) \subseteq \pi_W(a)^*, \quad (2.2)$$

so $\pi_W(a)$ is closable. It extends boundedly to $\mathcal{H}_W$ if and only if there is a constant C_a such that

$$W(b^*a^*ab) \leq C_a^2 W(b^*b) \quad (b \in \mathfrak{A}). \quad (2.3)$$

The least such C_a is the norm of the extension.

Proof. Positivity and polarization make $\langle \cdot, \cdot \rangle_W$ Hermitian and give Cauchy–Schwarz. If $b \in \mathcal{N}_W$, then

$$W((ab)^*ab) = \langle a^*ab, b \rangle_W = 0,$$

so $\mathcal{N}_W$ is a left ideal and (2.1) is well defined. For classes in D_W ,

$$\langle \pi_W(a)[b], [c] \rangle_W = W(c^*ab) = \langle [b], \pi_W(a^*)[c] \rangle_W.$$

This proves (2.2). The adjoint domain contains the dense subspace D_W , so $\pi_W(a)$ is closable. Finally,

$$\|\pi_W(a)[b]\|^2 = W(b^*a^*ab),$$

so the bounded-operator criterion is exactly (2.3). $\square$

For $a = a^*$, the operator $\pi_W(a)$ is symmetric and closable. Neither essential self-adjointness nor even the existence of a self-adjoint extension follows. For example,

$$P = -i \frac{d}{dx}, \quad \mathcal{D}(P) = C_c^\infty(0, \infty) \subset L^2(0, \infty),$$

is symmetric and closable, but its deficiency indices are $(1, 0)$, so it has no self-adjoint extension. Indeed, the adjoint equations $P^*u = iu$ and $P^*u = -iu$ give $u = ce^{-x}$ and $u = ce^x$, respectively. Only the first solution belongs to $L^2(0, \infty)$. Products of closures require a separate common-domain construction. We give that construction in Section 3.

Example 2.2 (An unbounded positive GNS multiplier). Let $\mathfrak{A} = \mathbb{C}[x]$, let $x^* = x$, and set

$$W(p) = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} p(t) e^{-t^2} dt.$$

Then W is positive, but multiplication by x is unbounded, since

$$\frac{\|\pi_W(x)[x^n]\|^2}{\|[x^n]\|^2} = \frac{\int_{\mathbb{R}} t^{2n+2} e^{-t^2} dt}{\int_{\mathbb{R}} t^{2n} e^{-t^2} dt} = n + \frac{1}{2}.$$

3 Closed forms and the common invariant domain

A finite value of one quadratic form permits one operator application. Repeated products require control of every algebra element. The distinction already occurs for a diagonal operator.

Example 3.1 (A maximal form domain need not be invariant). On $\ell^2(\mathbb{N})$, let $Ne_n = ne_n$, initially on $c_{00}(\mathbb{N})$. Its closure and the closed form $q(v) = \|Nv\|^2$ have domain

$$V = \left\{ v : \sum_{n \geq 1} n^2 |v_n|^2 < \infty \right\}.$$

Coordinate convergence proves that the maximal diagonal operator is closed. Finite coordinate truncations approximate every vector in its graph norm, so they prove the asserted closure. The vector $v_n = n^{-2}$ belongs to V , but $Nv \notin V$, since $\sum n^4 |v_n|^2 = \infty$. For the polynomial algebra generated by N , the common domain is instead

$$V_\infty = \bigcap_{k \geq 0} \left\{ v : \sum_{n \geq 1} n^{2k} |v_n|^2 < \infty \right\}. \quad (3.1)$$

Multiplication by each polynomial preserves V_∞ . Truncations converge in every displayed seminorm.

Return to Theorem 2.1. Write $D = D_W$, $H = \mathcal{H}_W$, and $A_a = \overline{\pi_W(a)}$ for $a \in \mathfrak{A}$. The graph topology on D is generated by $\|\xi\|$ and $\|\pi_W(a)\xi\|$, for all $a \in \mathfrak{A}$. It is a locally convex topology. Completion can require nets when $\mathfrak{A}$ has no countable family of controlling seminorms.

Theorem 3.2 (The exact common graph domain). *The completion of D in its graph topology identifies with*

$$D_W^{\text{cl}} = \bigcap_{a \in \mathfrak{A}} \mathcal{D}(A_a) \subset H. \quad (3.2)$$

On this domain, the maps $\pi_W^{\text{cl}}(a) = A_a|_{D_W^{\text{cl}}}$ form an invariant $*$ -representation. In particular,

$$\pi_W^{\text{cl}}(a)\pi_W^{\text{cl}}(b)\xi = \pi_W^{\text{cl}}(ab)\xi, \quad \langle \pi_W^{\text{cl}}(a)\xi, \eta \rangle = \langle \xi, \pi_W^{\text{cl}}(a^*)\eta \rangle. \quad (3.3)$$

Each action is continuous in the graph topology. For each a , the nonnegative form

$$q_a([b]) = W(b^* a^* ab) = \|\pi_W(a)[b]\|^2$$

is closable on H . Its closure is $\bar{q}_a(\xi) = \|A_a \xi\|^2$ on $\mathcal{D}(A_a)$. Thus (3.2) is exactly the intersection of these closed form domains. Each $\pi_W^{\text{cl}}(a)$ has closure A_a .

Proof. A graph Cauchy net (ξ_i) has a Hilbert limit ξ and limits $\eta_a = \lim_i \pi_W(a)\xi_i$. Closedness of A_a gives $\xi \in \mathcal{D}(A_a)$ and $A_a \xi = \eta_a$. If $\xi = 0$, closability gives $\eta_a = 0$ for every a . Hence the graph completion injects into the right side of (3.2).

To prove the reverse inclusion, fix ξ in that intersection and a finite set $F \subset \mathfrak{A}$. Set $c = \sum_{a \in F} a^* a$. For $u \in D$, the adjoint identity gives the estimate

$$\sum_{a \in F} \|\pi_W(a)u\|^2 = \langle \pi_W(c)u, u \rangle \leq \|\pi_W(c)u\| \|u\|. \quad (3.4)$$

Since $\xi \in \mathcal{D}(A_c)$, choose $u_j \in D$ with $u_j \rightarrow \xi$ and $\pi_W(c)u_j \rightarrow A_c \xi$. Apply (3.4) to $u_j - u_k$. Every $\pi_W(a)u_j$, $a \in F$, is Cauchy. Closedness identifies its limit with $A_a \xi$. Thus D approximates ξ in any prescribed finite collection of graph seminorms. Directing these approximations by finite sets and positive tolerances produces one net converging in all graph seminorms.

Let this net be (ξ_i) and fix $b \in \mathfrak{A}$. Then $\pi_W(b)\xi_i \rightarrow A_b \xi$ and $\pi_W(a)\pi_W(b)\xi_i = \pi_W(ab)\xi_i \rightarrow A_{ab}\xi$ for every a . Closedness gives $A_b \xi \in D_W^{\text{cl}}$ and $A_a A_b \xi = A_{ab}\xi$. Linearity follows by the same approximation. Approximating both vectors in the algebraic adjoint identity proves the second equality in (3.3). Continuity follows from $\|A_a A_b \xi\| = \|A_{ab}\xi\|$ and $\|A_b \xi\|$ itself.

The form norm for q_a is $(\|\xi\|^2 + \|\pi_W(a)\xi\|^2)^{1/2}$. Its completion is the graph of A_a , which proves the form assertion. Finally $\pi_W(a) \subseteq \pi_W^{\text{cl}}(a) \subseteq A_a$, so taking closures proves the last statement. Any closed $*$ -representation extending π_W contains the limits of these graph Cauchy nets and agrees with their multiplier limits. Thus it extends π_W^{cl} , which proves minimality. $\square$

This is the minimal closed representation in the sense of [4, Definition 2.4 and Lemma 2.6]. The word “closed” refers to completeness in the common graph topology. It does not assert that each restriction to D_W^{cl} is a closed operator or is self-adjoint. Example 3.1 separates the common domain from the maximal domain of one multiplier.

4 Dagger transfer and completion

Theorem 4.1 (Dagger transfer). *Let $T : \mathfrak{A} \rightarrow \mathfrak{B}$ be a $*$ -homomorphism of complex $*$ -algebras, let $\omega : \mathfrak{B} \rightarrow \mathbb{C}$ be positive, and put $W = \omega \circ T$. Then*

$$\langle a, c \rangle_W = \langle T(a), T(c) \rangle_\omega, \quad \mathcal{N}_W = T^{-1}(\mathcal{N}_\omega). \quad (4.1)$$

Consequently

$$U_0 : D_W \longrightarrow D_\omega, \quad [a] \longmapsto [T(a)], \quad (4.2)$$

is an isometric injection and extends uniquely to an isometry

$$U : \mathcal{H}_W \longrightarrow \mathcal{H}_\omega \quad (4.3)$$

with closed range. On the algebraic common domain,

$$U_0 \pi_W(a) = \pi_\omega(T(a)) U_0. \quad (4.4)$$

If $\mathfrak{B}$ is a C^* -algebra, both sides of (4.4) extend boundedly and

$$\|\pi_\omega(b)\| \leq \|b\|, \quad \|\pi_W(a)\| \leq \|T(a)\|, \quad (4.5)$$

while

$$U \pi_W(a) = \pi_\omega(T(a)) U \quad \text{on } \mathcal{H}_W. \quad (4.6)$$

Proof. For $a, c \in \mathfrak{A}$,

$$W(c^*a) = \omega(T(c)^*T(a)),$$

which proves positivity, (4.1), and the isometry (4.2). Every isometry of pre-Hilbert spaces extends uniquely to their completions; its range is closed because its domain is complete. Multiplicativity of T gives (4.4).

Suppose now that $\mathfrak{B}$ is a C^* -algebra. The C^* -order inequality gives

$$\omega(d^*b^*bd) \leq \|b\|^2 \omega(d^*d),$$

which proves the first estimate in (4.5). Taking $b = T(a)$ and $d = T(c)$ proves the second. Continuity then extends (4.4) to (4.6). No equality between arbitrary unbounded closure domains is used. $\square$

Proposition 4.2 (Transfer of closed graph domains). *Use the data of Theorem 4.1, and set $K = U\mathcal{H}_W \subset \mathcal{H}_\omega$ and $E = U_0D_W$. On the dense subspace $E \subset K$, define*

$$\rho(a) = \pi_\omega(T(a))|_E.$$

Then ρ is a $*$ -representation, and

$$UD_W^{\text{cl}} = D_\rho^{\text{cl}}, \quad U\pi_W^{\text{cl}}(a)\xi = \rho^{\text{cl}}(a)U\xi. \quad (4.7)$$

Here every closure defining D_ρ^{cl} is taken in K . For the ambient closures on $\mathcal{H}_\omega$ one has

$$U\mathcal{D}(\overline{\pi_W(a)}) \subseteq \mathcal{D}(\overline{\pi_\omega(T(a))}), \quad \overline{\pi_\omega(T(a))}U\xi = U\overline{\pi_W(a)}\xi. \quad (4.8)$$

If T is surjective, U is unitary and $UD_W^{\text{cl}} = D_\omega^{\text{cl}}$. Without surjectivity this last domain equality need not hold, even when U is unitary.

Proof. The algebraic intertwining identifies $\rho(a)$ with $U\pi_W(a)U^{-1}$ on E . As a map onto K , U is unitary. It identifies the individual graph closures and all graph seminorms. Theorem 3.2 therefore gives (4.7). If $\xi_j \in D_W$ approaches ξ in the graph norm of $\pi_W(a)$, its images approach $U\xi$ and have multiplier images approaching $U\overline{\pi_W(a)}\xi$. This proves (4.8). For surjective T , one has $U_0D_W = D_\omega$ and $T(\mathfrak{A}) = \mathfrak{B}$. Hence the Hilbert spaces and their full graph topologies agree. $\square$

Example 4.3 (A Hilbert isometry can miss the full target domain). Let $\mathfrak{A} = c_{00}(\mathbb{N})$ and $\mathfrak{B} = \mathbb{C}[n] + c_{00}(\mathbb{N})$, with pointwise multiplication and conjugation. Here $\mathbb{C}[n]$ denotes polynomial sequences on the positive integers. Let T be inclusion and set

$$\omega(b) = \sum_{n \geq 1} e^{-n} b_n, \quad W = \omega|_{\mathfrak{A}}.$$

All sums converge, and both functionals are positive and faithful. Both GNS completions identify with $H = \ell^2(\mathbb{N}, e^{-n})$, because they contain the finite sequences. Thus U is the identity on H . Every $\mathfrak{A}$ -multiplier is bounded, so $D_W^{\text{cl}} = H$. The element $n \in \mathfrak{B}$ acts by an unbounded multiplier. Finite truncations give cores for every polynomial multiplier. Therefore

$$D_\omega^{\text{cl}} = \left\{ v : \sum_{n \geq 1} e^{-n} n^{2k} |v_n|^2 < \infty \text{ for all } k \geq 0 \right\}.$$

The vector $v_n = e^{n/2}/n$ belongs to H , but fails the condition for $k = 1$. Thus $UD_W^{\text{cl}} \not\subseteq D_\omega^{\text{cl}}$. Only the restricted algebra $T(\mathfrak{A})$ appears in (4.7).

For a field interpretation, algebra products represent repeated operations on a common domain. Theorem 3.2 supplies this domain after positivity has supplied the Hilbert norm. Proposition 4.2 preserves it for the represented algebra. A vacuum vector, self-adjoint generators, locality, and a positive energy condition require their own constructions.

5 Positive unitization for nonunital C^* -algebras

The unit missing from a convolution algebra cannot be adjoined to an arbitrary positive functional without an additional bound. For a nonunital $*$ -algebra $\mathfrak{A}$, write $\mathfrak{A}^+ = \mathfrak{A} \oplus \mathbb{C}1$ and

$$\omega_m(a + \lambda 1) = \omega(a) + m\lambda. \quad (5.1)$$

Direct minimization in λ gives the algebraic criterion

$$\omega_m \text{ is positive} \iff \begin{cases} m \in \mathbb{R}, & m \geq 0, \\ \omega(a^*) = \overline{\omega(a)}, \\ |\omega(a)|^2 \leq m \omega(a^*a) & (a \in \mathfrak{A}). \end{cases} \quad (5.2)$$

Theorem 5.1 (The nonunital C^* case). *Let $\mathfrak{A}$ be a nonunital C^* -algebra and let $\omega : \mathfrak{A} \rightarrow \mathbb{C}$ be positive. Then ω is Hermitian and bounded, and the functional (5.1) on the standard unitization is positive exactly when*

$$m \in \mathbb{R}, \quad m \geq \|\omega\|. \quad (5.3)$$

Let (e_i) be any positive contractive two-sided approximate identity. For the minimal extension $m = \|\omega\|$, its GNS vector $\Omega = [1]$ satisfies

$$[e_i] \longrightarrow \Omega, \quad (5.4)$$

and the restricted representation of $\mathfrak{A}$ is bounded, nondegenerate, and cyclic:

$$\omega(a) = \langle \pi_\omega(a)\Omega, \Omega \rangle, \quad \overline{\pi_\omega(\mathfrak{A})\Omega} = \mathcal{H}_\omega. \quad (5.5)$$

For $m > \|\omega\|$, the vector $[1]$ of that larger extension is not the limit in (5.4).

Proof. Every self-adjoint element is the difference of two positive elements, so positivity makes ω Hermitian. If ω were unbounded, its values would be unbounded on the positive unit ball: otherwise the decomposition of an arbitrary element into four positive parts would bound it. One could then choose $a_n \geq 0$ with

$$\|a_n\| \leq 2^{-n}, \quad \omega(a_n) = 1.$$

The norm-convergent positive sum $a = \sum_n a_n$ would satisfy $\omega(a) \geq \sum_{n=1}^N \omega(a_n) = N$ for every N , a contradiction. Hence ω is bounded.

Positive contractive approximate identities exist. Cauchy–Schwarz and $e_i a \rightarrow a$ give

$$|\omega(a)|^2 = \lim_i |\omega(e_i a)|^2 \leq \|\omega\| \omega(a^* a). \quad (5.6)$$

Equations (5.2) and (5.6) prove positivity for $m = \|\omega\|$; adding $(m - \|\omega\|)$ times the positive quotient character $a + \lambda 1 \mapsto \lambda$ proves sufficiency for every larger m . Conversely, a positive functional on a unital C^* -algebra has norm equal to its value at the unit, so restriction of ω_m gives $\|\omega\| \leq m$.

It remains to identify the vector $[1]$. If $\|\omega\| > 0$, choose $\|a\| \leq 1$ with $|\omega(a)|$ arbitrarily close to $\|\omega\|$. Using $e_i a \rightarrow a$ and Cauchy–Schwarz shows

$$\omega(e_i^2) \longrightarrow \|\omega\|.$$

Since $0 \leq e_i^2 \leq e_i \leq 1$ in the unitization, also $\omega(e_i) \rightarrow \|\omega\|$. Therefore

$$\|[1] - [e_i]\|^2 = \|\omega\| - 2\omega(e_i) + \omega(e_i^2) \longrightarrow 0,$$

which is (5.4); the zero functional is immediate. Moreover $\pi_\omega(a)\Omega = [a]$, proving cyclicity, and $\pi_\omega(e_i)[a] = [e_i a] \rightarrow [a]$, proving nondegeneracy. The C^* -estimate in (4.5) proves boundedness. If $m > \|\omega\|$, the same computation instead gives

$$\lim_i \|[1] - [e_i]\|^2 = m - \|\omega\| > 0.$$

□

Segal's original formulation defines states in the possibly nonunital case on p. 75, states Theorem 1 and constructs an approximate identity on pp. 76–77, proves the quotient and multiplier bound on p. 78, and obtains the cyclic vector on pp. 78–79 of [3].

Example 5.2 (Positive unitization can fail algebraically). Let $\mathfrak{A} = c_{00}(\mathbb{N})$ with pointwise multiplication and conjugation, and define

$$\omega(a) = \sum_{n \geq 1} na_n.$$

This is a Hermitian positive functional. If a positive extension ω_m existed, Cauchy–Schwarz in the unitization, applied to the n th point mass e_n , would give

$$n^2 = |\omega_m(e_n)|^2 \leq \omega_m(1)\omega_m(e_n^*e_n) = mn$$

for every n , which is impossible for finite m .

6 The Gaussian obstruction to finite mass

For the full Weil functional, failure of positive unitization follows from its arithmetic local terms. It does not require a hypothesis about the zeros. The relevant functions approach the convolution identity while remaining inside $\mathcal{A}_\zeta$:

$$g_a(u) = \frac{e^{-u^2/(4a)}}{\sqrt{4\pi a}}, \quad \widehat{g}_a(t) = e^{-at^2}, \quad g_a^* \star g_a = g_{2a}, \quad a > 0. \quad (6.1)$$

Theorem 6.1 (No finite positive unitization of the Weil functional). *As $a \downarrow 0$, the functional in (1.6) satisfies*

$$L_\zeta(g_a) = \frac{a^{-1/2} \log(1/a)}{4\sqrt{\pi}} + O(a^{-1/2}). \quad (6.2)$$

In particular $L_\zeta(g_{2a}) > 0$ for all sufficiently small a , and

$$\frac{|L_\zeta(g_a)|^2}{L_\zeta(g_a^* \star g_a)} \sim \frac{\sqrt{2}}{4\sqrt{\pi}} a^{-1/2} \log(1/a) \longrightarrow \infty. \quad (6.3)$$

There is no positive extension of L_ζ to $\mathcal{A}_\zeta^+$ with a finite value at the adjointed unit.

Proof. Every derivative of g_a is a polynomial times a Gaussian. Thus $g_a \in \mathcal{A}_\zeta$. Completing the square proves (6.1), with the Fourier normalization (1.2). The polar contribution is $2e^{a/4} = O(1)$.

For $0 < a \leq 1$, split the Gaussian exponent into two equal parts. Since $\Lambda_{\text{vM}}(n) \leq \log n$, the absolute prime contribution is bounded by

$$\frac{e^{-(\log 2)^2/(8a)}}{\sqrt{\pi a}} \sum_{n \geq 2} \frac{\log n}{\sqrt{n}} e^{-(\log n)^2/8}. \quad (6.4)$$

The series is finite: its terms are at most n^{-2} for all sufficiently large n . Hence this contribution is exponentially small relative to $a^{-1/2}$.

Put $k_\infty(t) = \operatorname{Re} \psi(1/4 + it/2) - \log \pi$. The digamma expansion on a closed sector away from the negative real axis gives $\psi(z) = \log z - (2z)^{-1} + O(|z|^{-2})$ [1, Eq. 5.11.2]. Consequently $k_\infty(t) - \log |t|$ is bounded for $|t| \geq 1$. It is integrable on $[-1, 1]$, since k_∞ is smooth there. Therefore

$$\int_{\mathbb{R}} e^{-at^2} (k_\infty(t) - \log |t|) dt = O(a^{-1/2}).$$

Substitution $s = \sqrt{a} t$ gives

$$\frac{1}{2\pi} \int_{\mathbb{R}} e^{-at^2} \log |t| dt = \frac{a^{-1/2}}{2\pi} \left(\frac{\sqrt{\pi}}{2} \log(1/a) + \int_{\mathbb{R}} e^{-s^2} \log |s| ds \right).$$

The remaining integral is finite. Adding all three local terms proves (6.2). Applying that formula at a and $2a$ proves (6.3). A positive unitization of mass m would require $|L_\zeta(g_a)|^2 \leq mL_\zeta(g_{2a})$ by (5.2), which contradicts the divergence. $\square$

Theorem 6.2 (Positive finite Gaussian subspaces). *Fix distinct positive real numbers $\lambda_1, \dots, \lambda_r$. There exists $a_0 > 0$ such that the Weil form is positive definite on*

$$E_a = \operatorname{span}_{\mathbb{C}}\{g_{a\lambda_1}, \dots, g_{a\lambda_r}\} \quad (0 < a < a_0).$$

The dimension of E_a is r . This assertion is unconditional. The threshold a_0 may depend on the entire finite family.

Proof. The Gram matrix of the displayed functions is the real symmetric matrix

$$G(a)_{jk} = L_\zeta(g_{a(\lambda_j + \lambda_k)}).$$

Apply (6.2) at each of the finitely many scales $a(\lambda_j + \lambda_k)$. In matrix norm,

$$\frac{4\sqrt{\pi}\sqrt{a}}{\log(1/a)} G(a) \longrightarrow M, \quad M_{jk} = \frac{1}{\sqrt{\lambda_j + \lambda_k}}. \quad (6.5)$$

For $c \in \mathbb{C}^r$,

$$c^* M c = \frac{1}{\sqrt{\pi}} \int_{\mathbb{R}} \left| \sum_{j=1}^r c_j e^{-\lambda_j t^2} \right|^2 dt. \quad (6.6)$$

If this integral vanishes, continuity makes its integrand zero everywhere. Order the λ_j increasingly, divide the resulting identity by $e^{-\lambda_1 t^2}$, and let $t \rightarrow +\infty$. This gives $c_1 = 0$. Repeating proves $c = 0$. Thus M is positive definite. Let $\mu > 0$ be its smallest eigenvalue. Matrix convergence makes the left matrix in (6.5) differ from M by less than $\mu/2$ for all sufficiently small a . Hence $G(a)$ is positive definite. The same linear-independence argument applied to the Fourier transforms $e^{-a\lambda_j t^2}$ proves $\dim E_a = r$. $\square$

For $\lambda_1 = 1$ and $\lambda_2 = 4$, the limiting matrix is

$$M = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{5} \\ 1/\sqrt{5} & 1/\sqrt{8} \end{pmatrix}, \quad \det M = \frac{1}{20} > 0.$$

This gives a positive two-dimensional family directly from the local terms. The quantifiers in Theorem 6.2 fix the finite family before choosing its scale. They do not imply positivity on all of $\mathcal{A}_\zeta$ at once.

Corollary 6.3 (Excluded positive carriers). *There is no representation*

$$L_\zeta(g) = \langle \pi(g)\Omega, \Omega \rangle$$

on a positive Hilbert space with Ω of finite norm in a common invariant $*$ -representation domain. There is also no $*$ -homomorphism $T : \mathcal{A}_\zeta \rightarrow \mathfrak{B}$ to a C^* -algebra with a positive linear functional $\omega : \mathfrak{B} \rightarrow \mathbb{C}$ such that $L_\zeta = \omega \circ T$.

Proof. In the vector case, the adjoint identity on the invariant domain gives

$$|L_\zeta(g)|^2 \leq \|\Omega\|^2 \|\pi(g)\Omega\|^2 = \|\Omega\|^2 L_\zeta(g^* \star g).$$

This contradicts (6.3). In the C^* case, positivity implies boundedness and $|\omega(b)|^2 \leq \|\omega\| \omega(b^*b)$ by Theorem 5.1, or its unital Cauchy–Schwarz proof. Substitute $b = T(g)$ and use the same divergence. $\square$

This obstruction concerns the functional’s representing vector. It does not prevent bounded GNS multipliers. Indeed, under RH, (1.11) and $|\hat{f}(t)| \leq \|f\|_{L^1}$ give

$$L_\zeta((f \star b)^* \star (f \star b)) \leq \|f\|_{L^1}^2 L_\zeta(b^* \star b).$$

Thus every convolution multiplier is bounded under that hypothesis, and $D_{L_\zeta}^{\text{cl}} = \mathcal{H}_{L_\zeta}$. The functional itself has no finite representing vector. A positive functional on a proper test algebra, or a positive weight finite on a specified test ideal, remains compatible with this restriction. The latter requires its domain and all products to be defined before a GNS or field interpretation can be asserted.

7 Why an indefinite quotient does not give positivity

The Hermitian part of the construction exists without positivity. For a complex $*$ -algebra $\mathfrak{A}$ and a Hermitian functional W , put

$$B_W(f, g) = W(g^* f), \quad R_W = \{f : B_W(f, g) = 0 \text{ for every } g \in \mathfrak{A}\}.$$

Then R_W is a left ideal, since

$$B_W(af, g) = W(g^* af) = B_W(f, a^* g) = 0 \quad (f \in R_W).$$

The quotient $\mathfrak{A}/R_W$ consequently has a nondegenerate Hermitian pairing and an algebraic left action satisfying $B_W(af, g) = B_W(f, a^* g)$. This applies unconditionally to $(\mathcal{A}_\zeta, L_\zeta)$ by (1.8). It supplies neither a positive norm nor a class of the missing unit. Under positivity, Cauchy–Schwarz identifies R_W with $\mathcal{N}_W$, and Theorem 2.1 applies.

Let $V = \mathbb{C}^2$, $J = \text{diag}(1, -1)$, and

$$[x, y] = y^* J x.$$

The vectors $e_1 + e_2$ and $e_1 - e_2$ are isotropic, while their sum is not. Thus the isotropic cone $\{x : [x, x] = 0\}$ is not a linear subspace and cannot be divided out. The radical $\{x : [x, y] = 0 \forall y\}$ is zero here. In general, quotienting by the radical produces only a nondegenerate Hermitian space; it supplies neither a fundamental symmetry nor a complete Hilbert majorant. In this

finite-dimensional example such data can be chosen, but they are extra data, not consequences of the radical quotient.

Nor is a vector functional for an indefinite adjoint positive. On $M_2(\mathbb{C})$ put $A^\sharp = JA^*J$, let $\Omega = e_1$, and set $\omega(A) = [A\Omega, \Omega]$. For

$$X = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

one has

$$\omega(X^\sharp X) = [X\Omega, X\Omega] = [e_2, e_2] = -1. \quad (7.1)$$

Hence an indefinite pairing does not provide the positivity required in (1.10).

8 The independent arithmetic question

Proposition 8.1 (Formal factorization). *For a linear functional W on a complex $*$ -algebra $\mathfrak{A}$, the following are equivalent:*

1. W is positive;
2. there are a complex $*$ -algebra $\mathfrak{B}$ and maps

$$\omega : \mathfrak{B} \longrightarrow \mathbb{C}, \quad T : \mathfrak{A} \longrightarrow \mathfrak{B},$$

where ω is positive, T is a $*$ -homomorphism, and $W = \omega \circ T$.

Proof. Dagger transfer proves (2) $\Rightarrow$ (1). For the converse take $\mathfrak{B} = \mathfrak{A}$, $\omega = W$, and T equal to the identity. $\square$

For $W = L_\zeta$, formal factorization is therefore equivalent to the positivity in (1.10), and hence to RH. The identity factorization, or a Hilbert space whose spectral data are prescribed from the zeta zeros, cannot establish that positivity independently.

Open Problem 8.2 (Independent dagger realization of L_ζ). Construct a complex $*$ -algebra $\mathfrak{B}$, a positive linear functional $\omega : \mathfrak{B} \rightarrow \mathbb{C}$, and a $*$ -homomorphism $T : \mathcal{A}_\zeta \rightarrow \mathfrak{B}$ such that

$$L_\zeta = \omega \circ T. \quad (8.1)$$

Derive the carrier, the map, and the positivity of ω from the polar, archimedean, and prime-power structures in (1.6), without using RH, the multiset of zeta zeros, or an assertion equivalent to (1.10). Prove (8.1) with all three local terms and the Fourier and measure conventions above. Corollary 6.3 excludes a positive functional on a C^* -algebra as the target. Any admissible positive functional ω must fail the finite-mass unitization bound on the image of T . Theorem 3.2 and Proposition 4.2 then specify the common invariant domain supplied by this factorization.

The resulting representation would still need an additional comparison to a spectral determinant. Equality of functionals does not preserve trace multiplicities. For example, on $\mathbb{C}$ the positive functional $W(z) = mz$, with integer $m \geq 2$, has a one-dimensional GNS space. A scalar λ consequently has determinant $s - \lambda$ there. The trace realization $z \mapsto zI_m$ on $\mathbb{C}^m$ has the same functional and determinant $(s - \lambda)^m$. Thus a determinant realization must identify the operator, its domain, and spectral multiplicities separately.

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